Intro to Cyber Security

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Agenda for today

- Why cyber security
- Difference between safety and cyber security
- examples on Cyber Security algorithms
- Symmetric vs Asymmetric keys Algorithms
- What is Encryption ?
- Stream Cipher vs Block Cipher
- Padding schemes for Block Cipher
- ENCRYPTION VS VERIFICATION
- Cyber security stack (Crypto stack)

Why Cyber Security

Today, cars contain up to 150 electronic control units and about 100 million lines of software code – four times more than a fighter jet –, projected to rise to 300 million lines of code by 2030.

ISO/SAE 21434 (Road vehicles – Cybersecurity Engineering) and UNECE WP.29 (Regulation on Cybersecurity) are examples for the regulations needed for the manufacturers to comply to.

automotive manufacturers must comply with the United Nations Economic Commission for Europe (UNECE) Regulation WP.29/R155

Functional Safety and Cyber Security

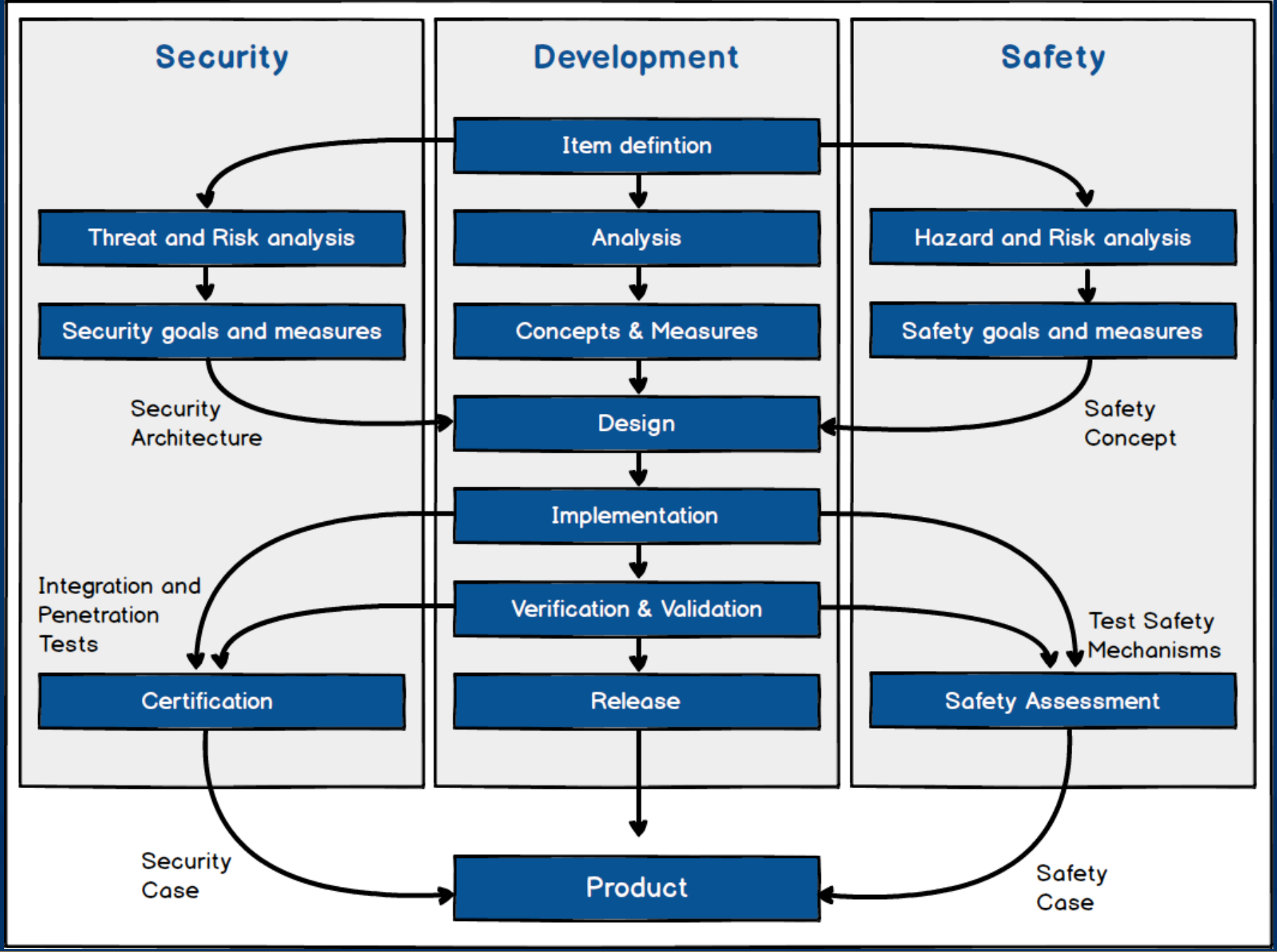
The difference between functional safety and cyber security is “calculated risk”

Compared to safety, **security** is a rather fast-moving discipline.

Especially due to the increase in networked components, security is increasingly coming into focus.

the danger increases steadily and previously successfully implemented countermeasures may no longer be sufficien

Safety is distinguished by a very static character.


Security
Development
Safety
Item defintion
Threat and Risk analysis
Analysis
Hazard and Risk analysis
Security goals and measures
Concepts & Measures
Safety goals and measures
Security Architecture
Design
Safety Concept
Implementation
Integration and Penetration Tests
Verification & Validation
Test Safety Mechanisms
Certification
Release
Safety Assessment
Security Case
Product
Safety Case

examples on Cyber Security algorithms

- Symmetric cryptography
  - AES
  - DES
- Asymmetric cryptography
  - RSA
  - ECC
- Hash-function
  - MD5
  - SHA-2

Key Differences	Symmetric Encryption	Asymmetric Encryption
Size of cipher text	Smaller cipher text compares to original plain text file.	Larger cipher text compares to original plain text file.
Data size	Used to transmit big data.	Used to transmit small data.
Resource Utilization	Symmetric key encryption works on low usage of resources.	Asymmetric encryption requires high consumption of resources.
Key Lengths	128 or 256-bit key size.	RSA 2048-bit or higher key size.
Security	Less secured due to use a single key for encryption.	Much safer as two keys are involved in encryption and decryption.
Number of keys	Symmetric Encryption uses a single key for encryption and decryption.	Asymmetric Encryption uses two keys for encryption and decryption
Techniques	It is an old technique.	It is a modern encryption technique.
Confidentiality	A single key for encryption and decryption has chances of key compromised.	Two keys separately made for encryption and decryption that removes the need to share a key.
Speed	Symmetric encryption is fast technique	Asymmetric encryption is slower in terms of speed.
Algorithms	RC4, AES, DES, 3DES, and QUAD.	RSA, Diffie-Hellman, ECC algorithms.

ENCRYPTION

This process converts the original representation of the information, known as plaintext, into an alternative form known as ciphertext.

To Encrypt a plaintext we need :

- ENCRYPTION ALGORITHM : AES , RSA
- DATA MODE : CBC , EBC

CRYPTOGRAPHY MODES

- BLOCK CIPHER
  - ECB
  - CBC
- STREAM CIPHER

ECB

Block cipher basics
Plaintext data
Block cipher encryption
Ciphertext
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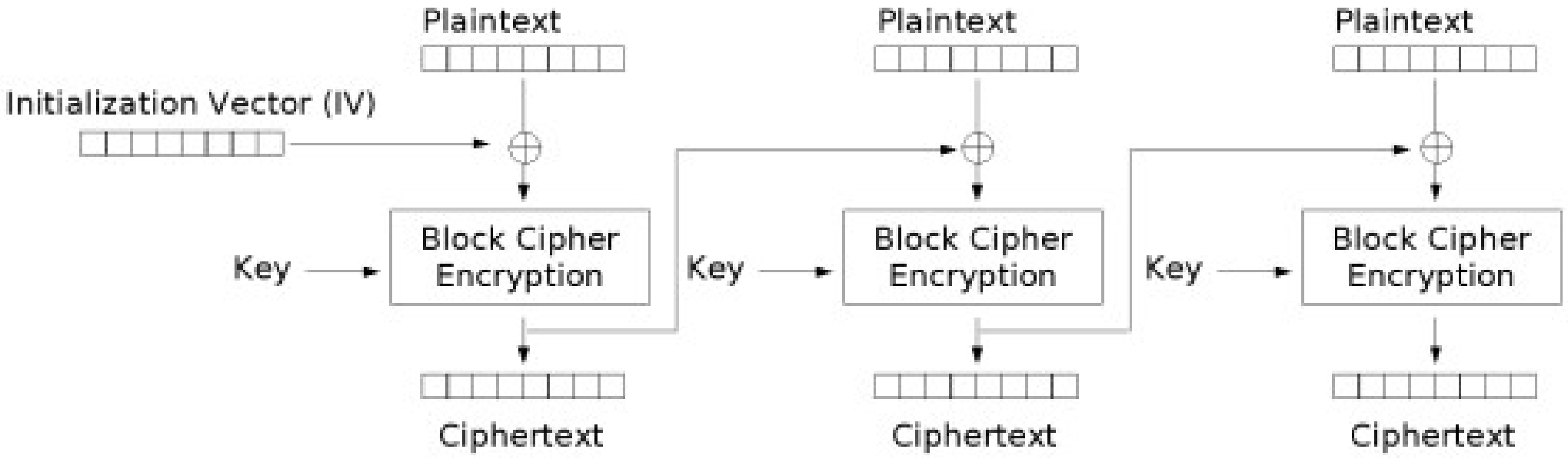
CBC



Cipher Block Chaining (CBC) mode encryption

Padding schemes for block ciphers

To perform encryption with a block cipher in ECB or CBC mode the length of the input to be encrypted *must* be an exact multiple of the block length B in bytes.

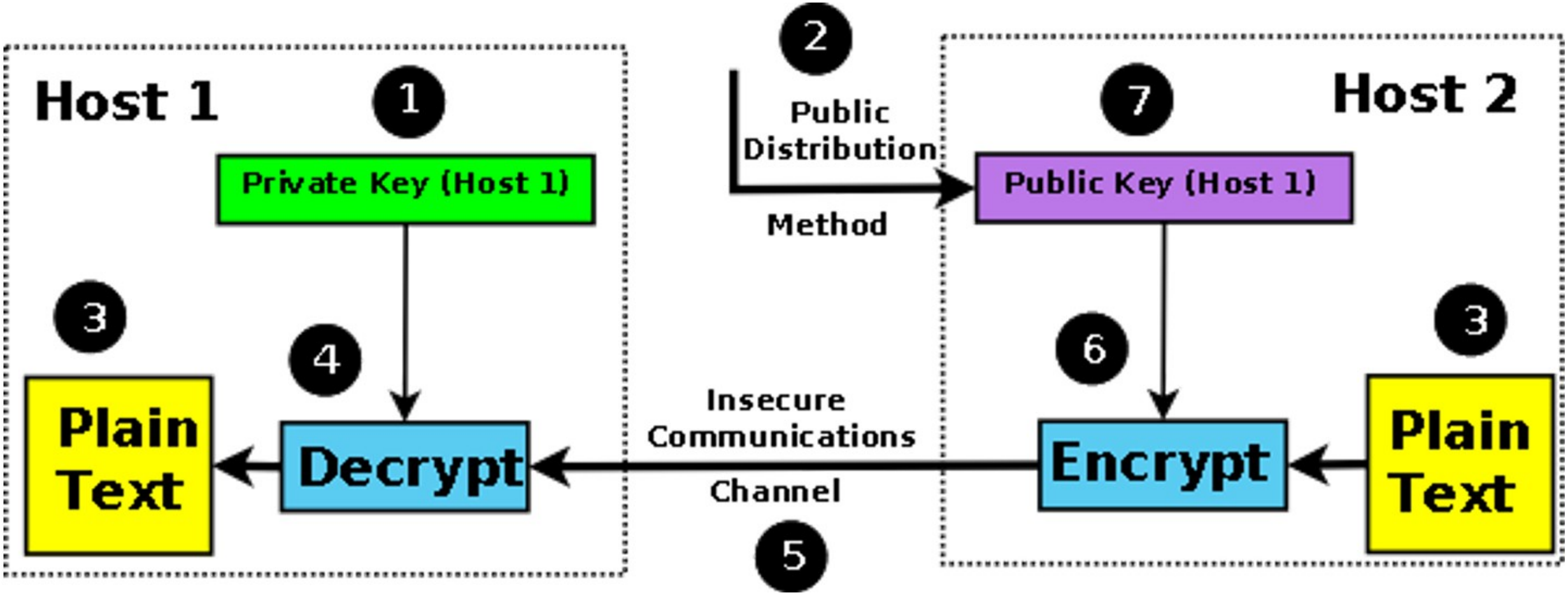
For Triple DES the block length B is 8 bytes (64 bits) and for all AES variants it is 16 bytes (128 bits).

If the length of the data to be encrypted is not an exact multiple of B, it must be padded to make it so.

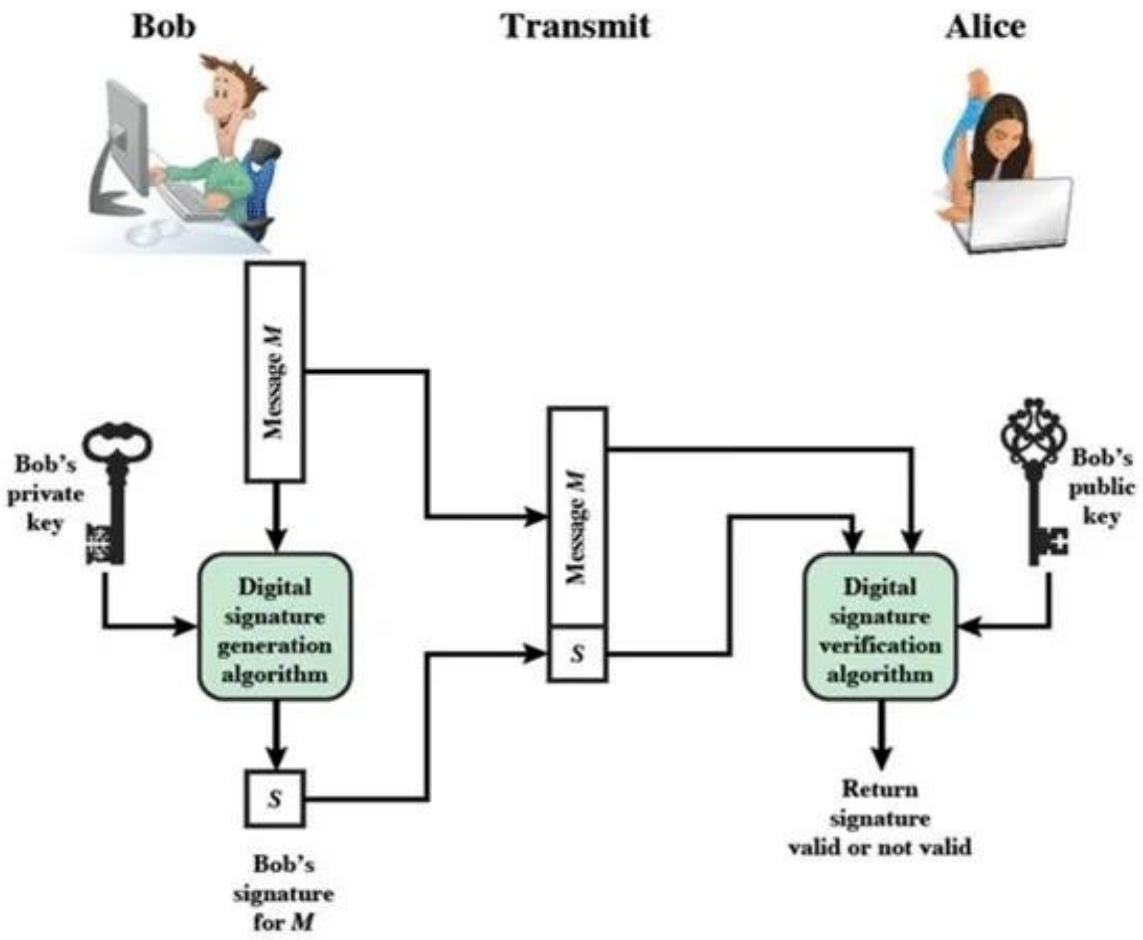
After decrypting, the padding needs to be removed.

The most popular is "PKCS5" padding also there is OneAndZeroes Padding.

ENCRYPTION


Host 1
1
Private Key (Host 1)
3
4
Plain Text
Decrypt
2
Public Distribution Method
Insecure Communications Channel
5
Host 2
7
Public Key (Host 1)
6
3
Encrypt
Plain Text

VERIFICATION AND AUTHENTICATION



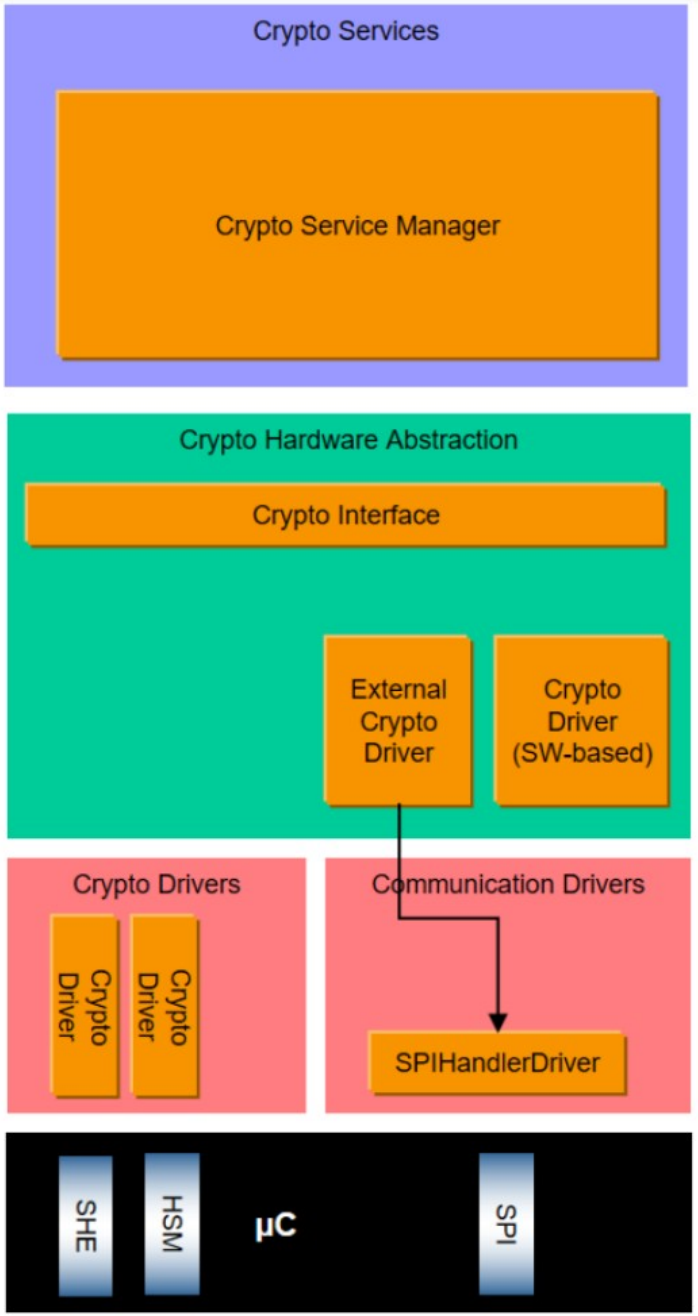
BREAK….

CRYPPTO STACK

CSM

CryIf

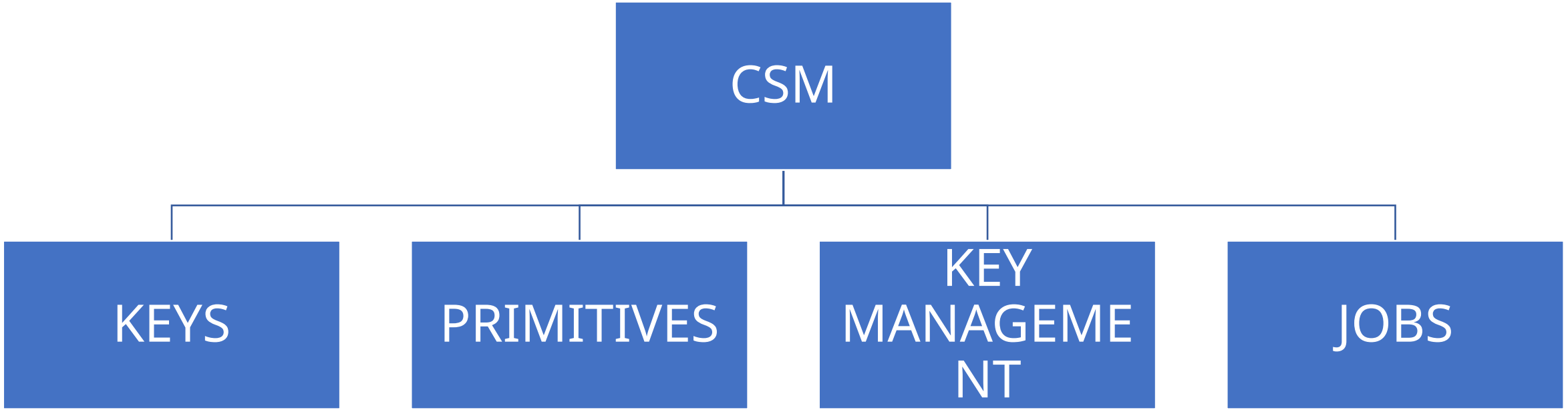
Crypto Drive :
-Software Library
-Hardware Library "HSM"

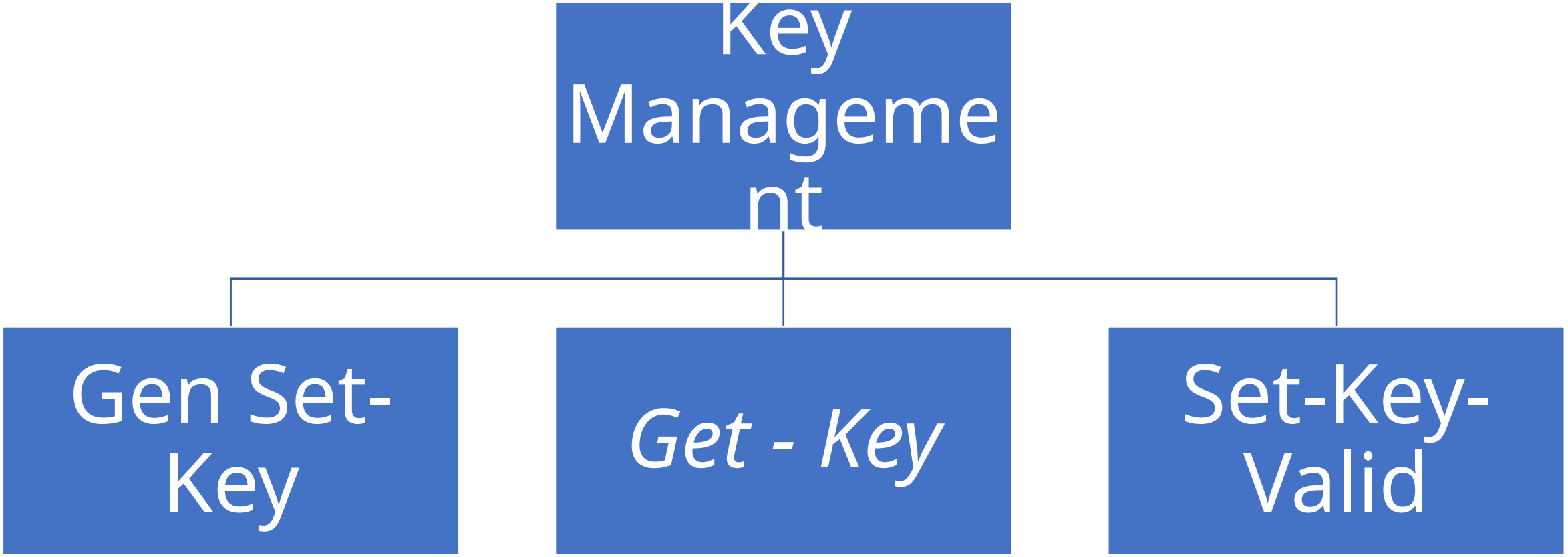


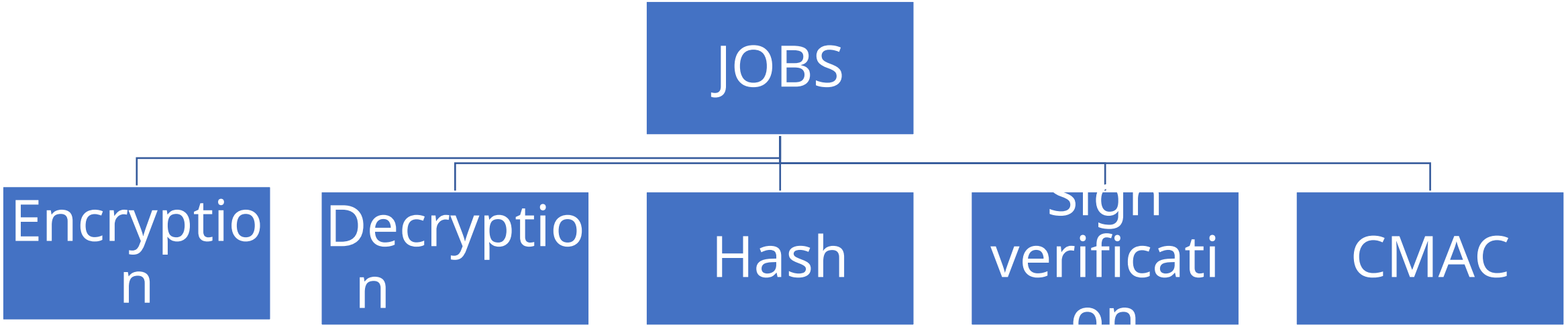
CSM
Crypto Service Manager

Crypto Service Manager (CSM) is a Service layer module which provides standard interface to the application to access basic cryptographic services provided by the Software/Hardware crypto driver.

These services can be accessed by application synchronously (services are completed when the function returns) or asynchronously (callback routine is called on completion of the service).


CSM
KEYS
PRIMITIVES
KEY MANAGEME NT
JOBS


Key Manageme nt
Gen Set-Key
Get - Key
Set-Key-Valid


JOBS
Encryption
Decryption
Hash
Sign verification
CMAC

CryIF Crypto Interface

Crypto Interface (CryIf) is an ECU abstraction layer module which abstracts the services provided by the low level Software or Hardware driver from CSM.

so that CSM can provide a standardized interface to the application.

It gives interfaces to be used by the application through the CSM

These interfaces are actually implemented
in the in the Crypto Driver (found in the MCAL layer of the Same ECU)
or
from External Crypto Driver (from another ECU)

Crypto Driver

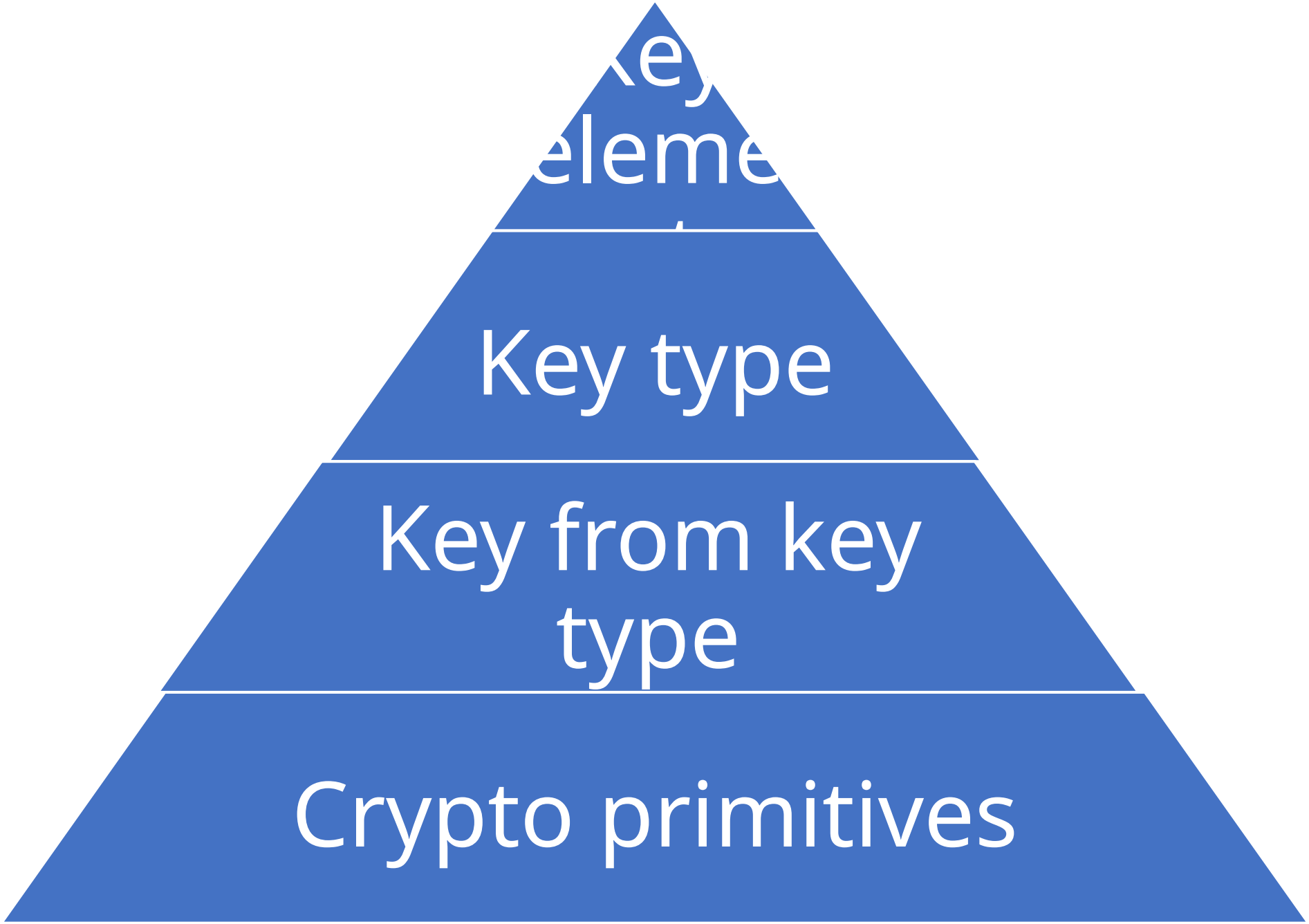
Crypto Driver is an MCAL module which has an implementation for a cryptographic service.

It can be software or a hardware module.

Hardware is also used to provide the functionality but it is then mostly accessed using an SPI handler.

Key elements

- key
- Initialization vector
- modulus
- Exponent


Key type
Key from key type
Crypto primitives

Thank You

“Nothing great was ever achieved without enthusiasm”

Ralph Waldo Emerson

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